



***B.Tech. Degree V Semester Supplementary Examination
April 2018***

**SE 15 - 1505 MANUFACTURING PROCESS
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Differentiate between carbon steel and stainless steel.
 (b) What is the main difference between a brass and a bronze?
 (c) Explain the purpose of electrode coating in arc welding.
 (d) What are the gases used for gas welding? Why does the welding torch get over heated while doing the gas welding operation?
 (e) Explain submerged arc welding.
 (f) What are the desirable qualities of moulding sand?
 (g) Sketch a mould cavity in a moulding flask and show the following:
 (i) Pouring basin (ii) Sprue (iii) Skim bob (iv) Riser
 (v) Core (vi) Runner (vii) Parting line (viii) Drag
 (ix) Cope.
 (h) Sketch any two types of rolling mill.
 (i) List the common cutting tool materials.
 (j) With the help of a neat sketch explain AJM.

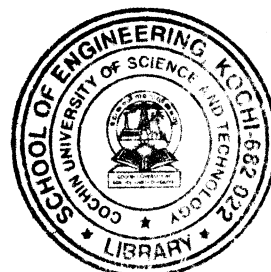
PART B

(4 × 10 = 40)

- II. (a) Distinguish between gray cast iron, white cast iron, malleable cast iron, and ductile cast iron. How is each produced? (7)
 (b) Write composition and use of the following alloys. (3)
 (i) Invar (ii) Elinvar (iii) Platinite.
 OR
 III. (a) What is the difference between Carburizing and Nitriding treatment? Up to what depth components can be hardened by carburizing process? (6)
 (b) Explain the process of annealing? (4)
 IV. (a) What are the common weld defects? (4)
 (b) Explain the following welding operations. (6)
 (i) Spot welding.
 (ii) Projection welding.
 (iii) Seam welding.

OR

(P.T.O.)



- V. (a) Describe electron beam welding. (6)
(b) How are welds inspected? (4)
- VI. (a) With the help of a neat sketch explain centrifugal casting. (6)
(b) How is slush casting done? (4)
- OR**
- VII. (a) List the different types of allowances provided on the pattern. Explain any three in detail. (5)
(b) List the possible defects in casting. Explain any three in detail (5)
- VIII. (a) List the advantages and limitations of hot working. (4)
(b) What is forging? Explain the different types of forging. (6)
- OR**
- IX. (a) Differentiate between orthogonal and oblique cutting. (3)
(b) Sketch the main parts of a Lathe as a block diagram and list the operations that can be performed on a lathe. (7)
